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Abstract

We study academic dishonesty using the standard tools of modern science: a two-strategy game, a control parameter, and a graph with dashed lines on it. Students choose between Honest and Cheat, with payoffs determined by course difficulty, peer effects, surveillance intensity, and the increasingly reasonable fear of being the only honest person in the room. Using replicator dynamics, we show that honesty is not generally a natural state of the system: when the first cheater can operate under the radar, the all-honest population fails to be an equilibrium. This observation is morally discouraging but mathematically convenient.

We then identify a bistable regime in which both widespread cheating and grudging compliance are self-reinforcing, separated by an unstable threshold of institutional panic. Beyond a critical surveillance level, the high-cheating regime loses stability and the population undergoes a discontinuous transition to what administrators may describe as an integrity culture and mathematicians may describe as the only attractor left. We derive conditions for this transition, provide numerical simulations, and discuss the broader policy lesson that virtue is frequently just equilibrium selection with better branding. We conclude by observing that fairness can also be restored, in principle, by ensuring that everyone cheats identically.

1 Introduction: Academic Integrity as a Dynamical-Systems Problem

Academic dishonesty has traditionally been studied by educators, ethicists, and, on take-home exams, by several students at once. In this paper we adopt the more mature approach of dynamical systems. We model a classroom as a population of strategic agents and ask a question of obvious scientific and administrative importance: under what conditions does a course transition from endemic misconduct to a state that the provost may safely call “integrity”?

Our central claim is that honesty is not, in general, a default property of academic life. If the first cheater faces negligible detection risk, then a fully honest population is vulnerable to invasion. In that sense, integrity is not a primitive moral fact but a stability property of a nonlinear system. This is bad news for philosophy, good news for bifurcation theory, and excellent news for people who already wanted to draw Figure 2.

The model exhibits the full lifecycle of modern educational policy. Under weak enforcement, cheating is stable. Under intermediate enforcement, the system is bistable: one may obtain either a mostly honest class or a morally lively one, depending on initial conditions and who already has the answer key. Under sufficiently strong enforcement, the high-cheating regime loses stability and the population is forced toward low cheating. We call this phenomenon catastrophic honesty.

Having summarized the problem at the level of intuition and mechanism, we now restate it in the conventional scholarly register. The next several paragraphs review prior work on academic dishonesty, deterrence, strategic interaction, and threshold effects before introducing the formal model developed in the remainder of the paper.

Academic dishonesty remains a persistent challenge in education, with surveys consistently reporting high rates of cheating among both undergraduate and graduate students [22, 21]. Such misconduct is not a harmless college phase: students who engage in academic cheating are more likely to behave unethically in professional and civic life [15, 18],

and widespread cheating erodes the trust essential to educational and scientific institutions. In research contexts, for example, studies have found troubling levels of falsification, plagiarism, and other misconduct, contributing to paper retractions and undermining scientific integrity [19, 13]. These observations underscore that academic fraud is both prevalent and consequential, warranting rigorous inquiry into its causes and potential remedies.

Researchers have approached academic dishonesty through multiple theoretical lenses. Criminological frameworks like the classic Fraud Triangle posit that fraudulent acts arise from a confluence of pressure, opportunity, and rationalization [9]. Economic deterrence models similarly predict that cheating will be rare only when the expected cost of being caught (e.g. probability of detection times penalty) outweighs the illicit benefit [4, 1]. In educational psychology, social-cognitive theories emphasize moral attitudes and norms: for instance, the Theory of Planned Behavior has been shown to predict students' cheating intentions based on their attitudes, perceived norms, and perceived control over cheating [3]. Empirical studies support elements of these frameworks – finding that students under high performance pressure or ample opportunity are more prone to cheat, while those with strong internalized values or intrinsic motivation tend to refrain [26, 20, 10]. Differences in individual ethics and maturity also play a role: students with greater ethical development or commitment to learning report lower tolerance for cheating [12, 25]. Together, these perspectives suggest that dishonesty is a multifaceted behavior driven by rational cost-benefit considerations, situational pressures, and psychological rationalizations.

Despite extensive prevention efforts, dishonest behavior has proven resilient, often adapting to circumnavigate new deterrents. Traditional “static” interventions (e.g. stricter proctoring or honor codes) sometimes yield only modest improvements as students find new ways to cheat, suggesting the strategic nature of the problem. Game-theoretic analyses indicate that cheating can remain a favorable strategy under certain conditions, making purely honest behavior difficult to sustain as a stable equilibrium [11]. In effect, there is an evolutionary arms race between cheaters and enforcers: as long as the perceived payoff from cheating exceeds the risks, a subset of individuals will continue to exploit the system. However, both theoretical models and real-world observations hint at a tipping-point dynamic – a nonlinearity in which incremental increases in oversight or penalties can eventually trigger an abrupt transition from a high-cheating regime to a low-cheating (high-compliance) regime. In other words, dishonest conduct may persist until enforcement or social pressure crosses a critical threshold, at which point the population's behavior “cheat-then-cease” suddenly shifts toward honesty. Evolutionary models of institutional misconduct support this possibility: for example, studies of tax evasion and corruption games have found multiple stable equilibria, with one equilibrium characterized by prevalent cheating and another by widespread compliance, depending on parameters like audit probability or fine severity [7, 8, 5, 2]. Likewise, in reputation-based systems, raising the standards or costs for misbehavior beyond a certain point can eliminate defection entirely, forcing a transition to full cooperation [24]. These phenomena are broadly consistent with threshold models of collective behavior, in which gradual changes in incentives or social norms can lead to sudden, discontinuous shifts in the population's overall behavior [14].

Against this backdrop, we develop an evolutionary game-theoretic model that captures the cheat-then-cease tipping dynamics. Our approach treats students' choices – to cheat or act honestly – as strategies in a population, and it explicitly incorporates an institutional enforcement parameter (e.g. rigor of proctoring or severity of punishment) that influences the payoff for cheating. Evolutionary game theory (EGT) provides a natural framework for this analysis, as it tracks how the frequency of strategies evolves under selective pressures and has been successfully applied to model the spread of behaviors in socio-economic contexts [16]. While prior works have used game-theoretic models to examine academic cheating and plagiarism (e.g. static payoff analyses of copying strategies) [17], our model innovates by focusing on the dynamic evolution of dishonest behavior and the bifurcations (qualitative regime changes) induced by enforcement. Using replicator dynamics, we analyze the stability of equilibria as enforcement parameters vary. Bifurcation analysis reveals critical enforcement thresholds that mark a phase transition: below the threshold, the system gravitates toward a mixed equilibrium with persistent cheaters, whereas above the threshold, the only stable equilibrium is one of near-universal honesty. In essence, the model formalizes how strengthening deterrence measures can nudge the system past a tipping point, abruptly extinguishing the incentive to cheat at the population level.

Our results offer a theoretical explanation for the observed cheat-then-cease phenomenon and provide insight into policy design for academic integrity. By identifying the conditions under which dishonest behavior is evolutionarily sustained versus eradicated, we contribute to both the behavioral literature on fraud deterrence and the methodological literature on applying EGT to human institutions. The analysis bridges micro-level decision mechanisms (individual cost-benefit calculations and moral factors) with macro-level outcomes (population-wide norm shifts), highlighting the importance of nonlinear effects in deterrence. In the following sections, we detail the model formulation (Section 2), analyze the evolutionary equilibria and their stability (Section 4), and discuss how the findings can inform more effective strategies to promote academic honesty (Section 5). Our approach demonstrates the value of marrying behavioral theory with dynamical systems analysis to better understand and ultimately mitigate academic dishonesty in the long run.

2 Model, Assumptions, and Other Ways to Lose Faith in Humanity

2.1 Game Setup

We consider a course with a large number of students who each choose one of two strategies: **Cheat** or **Honest**. If a student cheats, they potentially gain a higher grade or save effort (a payoff benefit), but risk incurring a penalty if caught. An honest student avoids penalty risk but may receive a lower grade if others are cheating and inflating the grade distribution. We denote x as the fraction of students who are cheating in the population. The parameters influencing payoffs are:

- **Class Difficulty (D):** Higher D increases the benefit of cheating (e.g. in a very hard class, copying answers greatly boosts one's grade).
- **Surveillance Intensity (S):** Represents how actively and effectively the instructor or institution monitors for cheating (higher S means a higher probability of catching cheaters, via proctors, plagiarism detectors, etc.).
- **Penalty Severity (K):** The institutional penalty if caught - this could be point deduction, failing grade, or harsher disciplinary action. A higher K means a caught cheater's payoff plummets more.
- **Peer Pressure Factor (P):** Captures the influence of peer behavior. If many classmates cheat, the social and academic pressure to also cheat increases. This can be interpreted as an increase in the perceived benefit of cheating when x is large (e.g. to "level the playing field" or because friends facilitate cheating).
- **Emotional Maturity (M):** A higher maturity (or ethical commitment) effectively reduces a student's propensity to cheat by increasing the internal cost (guilt, fear of long-term consequences) of cheating. For simplicity, we model maturity as reducing the net benefit a student perceives from cheating. By definition, this includes payoff from an intrinsic motivation to learn, the benefit gained from actually learning the material.

Bounds on each of these parameters are discussed in Appendix A.

2.2 Payoff Structure

We formulate payoff functions for each strategy, drawing from the rational-choice perspective. Let U_C denote the payoff (utility) for a student who cheats, and U_H for one who remains honest. We decompose U_C into expected benefits minus costs:

Benefit of cheating: We assume a cheating student gains an advantage equivalent to a certain grade boost or effort saved. We set this benefit as $B(D, x)$. The benefit likely increases with class difficulty D (cheating is more rewarding in a harder class) and also with the fraction of other cheaters x (if many are cheating, an honest student is at a relative disadvantage, so the effective benefit of cheating to "keep up" is higher). We can model $B(D, x) = D \cdot (1 + Pf(x))$, where $f(x)$ is an increasing function capturing peer influence (e.g. $f(x)$ could be linear $f(x) = x$ for peer normalization, or nonlinear for threshold effects). For tractability, we use a simple linear form in our simulations, $B(D, x) = D \cdot (1 + Px)$, which means peer cheating slightly amplifies the personal benefit of cheating (up to $2x$).

Cost of cheating: If a cheater is caught, they incur a penalty. Let $p(x, S)$ be the probability of getting caught, which depends on surveillance S and possibly on $x \in [0, 1]$ (the prevalence of cheating). The dependence on x models how detection becomes easier or harder as more students cheat. In a baseline scenario, one might assume more cheaters make detection easier (e.g. unusual score patterns or a greater chance someone slips up). We capture this by letting p increase with x . On the other hand, one could argue that if everyone cheats, it's harder to single out individuals - a "safety in numbers" effect - causing p to drop at extreme x . To allow rich dynamics, we adopt a model that combines both effects: low x yields low detection (a lone cheater can fly under the radar), moderate x yields higher detection (professor grows suspicious), and p may plateau or even decline if x is very high (the instructor is overwhelmed or lenient if the whole class misbehaves). A polynomial form

$$p(x, S) = S \cdot (x - cx^2), \quad c \in [0, 1]$$

serves as a convenient choice. This form gives $p = 0$ at $x = 0$ (no cheaters to catch) and at $x = 1$ (all cheaters) yields $p(1, S) = S(1 - c)$ - nonzero if $c < 1$, or exactly 0 if $c = 1$ for an extreme where a perfectly coordinated cheating conspiracy fools the system. This form of $p(x, S)$ is maximized when $x = \min(1, \frac{1}{2c})$, with

$$\max_x p(x, S) = \begin{cases} 1 - c & \text{for } c < \frac{1}{2} \\ \frac{1}{2c} - c(\frac{1}{2c})^2 = \frac{1}{4c} & \text{for } c \geq \frac{1}{2} \end{cases} \quad (1)$$

The coefficient c tunes the "safety in numbers" effect: if $c = 0$, p increases linearly with x (no safety in numbers at all); if c is close to 1, detection probability peaks at intermediate x and falls near $x = 1$. Precisely, $1 - c$ is the probability of

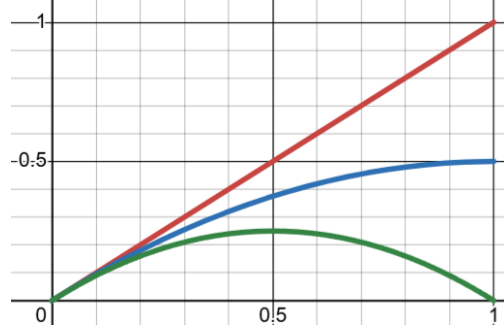


Figure 1: Cost of cheating $p(x, S) = S(x - cx^2)$, $S = 1$ for various values of c , the “safety in numbers” parameter. Top line: $c = 0$, mid: $c = 0.5$, bottom: $c = 1$. Intuitively, if there is no safety in numbers, the probability of getting caught increases linearly with the number of other people who are also cheating, and if there is high safety in numbers, probability of getting caught decreases if more than half the class is cheating. In this figure, where $S = 1$, the probability of getting caught if everyone is cheating is $1 - c$.

getting caught if everyone cheats (c is therefore the probability of not getting caught if everyone cheats). In our analysis we consider a representative case with $0 < c < 1$ so that cheating has diminishing returns in detection when ubiquitous. A visual overview of $p(x, S)$ for $S = 1$ in Figure 1.

Effective penalty: Given detection probability $p(x, S)$, the expected cost to a cheater is $p(x, S) \cdot K$, where K is the penalty severity. A highly mature student (high M) might subjectively amplify this cost (through fear or guilt), but we subsume such effects into a lower effective benefit or higher perceived K . For simplicity, we treat K as already reflecting the student’s risk tolerance (lower K for a risk-taking or immature student who doesn’t weigh future costs, higher K for a risk-averse or mature student who is terrified of getting caught).

Combining these, we can express the payoff difference between cheating and honesty. Let’s take the honest payoff U_H as a baseline (e.g. zero, representing the utility of studying diligently and receiving an honest grade). Then the net advantage of cheating is:

$$\Delta U = U_C - U_H = (B(D, x) - p(x, S) \cdot K - C_{moral}) - 0, \quad (2)$$

where C_{moral} is an optional term for moral cost (which we can fold into an adjustment of K or B for those with high M).

A rational student will choose to cheat if $\Delta U > 0$ (cheating yields higher expected payoff) and remain honest if $\Delta U < 0$. In equilibrium, all indifference conditions ($\Delta U = 0$) and best responses should be satisfied. We note here again that loss in content knowledge from cheating is folded into C_{moral} .

2.3 Equilibria, Including the Bad Ones

In a one-shot static game of N students, a Nash equilibrium occurs when each student’s strategy is a best response to the others. Because students are symmetric, we anticipate equilibria in which either (a) all students follow the same pure strategy, or (b) a certain fraction mix between cheating and honesty, yielding an effective population state $\hat{x}$. Due to the payoff externalities (peer pressure and detection depending on x), multiple equilibria are possible. Notably, if $\Delta U(x)$, viewed as a function of the cheating fraction, changes sign depending on x , the game can exhibit a coordination problem: cheating might be advantageous when many others cheat (so one is not caught and must cheat to keep up), but disadvantageous when few others cheat (because a lone cheater stands out to enforcers and gains little if the class is easy). This strategic complementarity can create two stable equilibria (one mostly cheating, one mostly honest) separated by an unstable tipping point. Such phenomena are well-known in coordination games and can be analyzed via best-response dynamics or evolutionary dynamics.

By Nash’s seminal theorem, at least one equilibrium (possibly in mixed strategies) is guaranteed to exist for this finite game [23]. For our two-strategy scenario, a mixed-strategy equilibrium would correspond to an internal population fraction $\hat{x}$ where $\Delta U(\hat{x}) = 0$, meaning students are indifferent between cheating and honesty when exactly $\hat{x}$ of their peers cheat. We can solve $\Delta U = 0$ to find candidate equilibria:

All-honest equilibrium ($x = 0$): This is an equilibrium if $\Delta U(0) < 0$, i.e. when no one is cheating, a student has no incentive to deviate and cheat. Plugging $x = 0$ into ΔU : $B(D, 0) - p(0, S)K = D \cdot (1 + P \cdot 0) - 0 \cdot K = D$. So

$\Delta U(0) = D$. Here we see a crucial point: if the class is even moderately difficult ($D > 0$), then $\Delta U(0) > 0$ - a lone student would gain by cheating when nobody else is cheating, because initially there is essentially no risk ($p(0, S) = 0$). This implies $x = 0$ is not a Nash equilibrium in a difficult class unless surveillance S is effectively infinite (catching even a first cheater with certainty). Intuitively, if no one else is cheating, one can cheat “under the radar” and benefit, so pure honesty is unstable. For an all-honest state to hold as equilibrium, we require either trivial class difficulty ($D = 0$, no benefit to cheating) or a policy that imposes a deterrent even on the first cheater (e.g. preemptive surveillance). In our model, we could incorporate a baseline detection probability $p_0 = S_0 > 0$ even at $x \approx 0$ to represent proactive enforcement (e.g. random checks). In that case, $\Delta U(0) = D - S_0 K$. A sufficiently large S_0 (such that $S_0 K > D$) could indeed make $x = 0$ an equilibrium by making the lone-cheater payoff negative. We will see shortly that this condition relates to the critical enforcement threshold for eliminating cheating.

All-cheat equilibrium ($x = 1$): This is an equilibrium if $\Delta U(1) \geq 0$, meaning when everyone cheats, no single student would gain by unilaterally switching to honest. If all others cheat, an honest student would avoid penalty risk but likely receive a much lower grade relative to cheaters (especially if grading is curved or competitive). In our model, $\Delta U(1) = B(D, 1) - p(1, S)K$. Using our forms: $B(D, 1) = D(1 + P)$ and $p(1, S) = S(1 - c)$ (with $c < 1$ giving some detection even when cheating is rampant). Thus $\Delta U(1) = D(1 + P) - S(1 - c)K$. If surveillance is weak (S small) or the “safety in numbers” effect is strong (c near 1, so $p(1) \approx 0$), then $\Delta U(1) > 0$ - cheating remains beneficial even in the all-cheat scenario, so $x = 1$ is a stable equilibrium. However, if enforcement is sufficiently high such that $S(1 - c)K > D(1 + P)$, then $\Delta U(1)$ becomes negative - meaning if the entire class cheats under such stringent enforcement, each cheater’s expected penalty would outweigh the benefit. In that case, an individual would actually prefer to not cheat given everyone else is cheating (a somewhat paradoxical situation indicating the all-cheat state is unstable). Practically, this would require extremely intense surveillance or punishment (e.g. if every cheater is almost certainly caught and heavily penalized, each student might rather take their chances studying). We identify this condition as the point where the cheating regime collapses: beyond this parameter threshold, the all-cheat equilibrium disappears.

Mixed (interior) equilibrium ($0 < \hat{x} < 1$): Solving $\Delta U(\hat{x}) = 0$ generally yields an interior equilibrium. Using a specific functional form for concreteness, suppose $p(x, S) = S(x - cx^2)$ as introduced. Setting $\Delta U = 0$: $D(1 + Px) - S(x - cx^2)K = 0$.

This quadratic equation can be solved for x . In the special case $P = 0$ (no peer benefit) and $c = 0$ (detection increases linearly with x), the equilibrium simplifies to $\hat{x} = \frac{D}{SK}$. More generally, with peer effects and nonlinear detection, multiple solutions can arise. The algebra yields:

$$cSKx^2 - SKx + D(1 + Px) = 0, \quad (3)$$

which can have 0, 1, or 2 positive solutions in $(0, 1)$ depending on parameters. When two interior solutions exist, the lower- x solution typically is stable (attracting dynamics towards more honest behavior) and the higher- x solution is unstable (a tipping point separating basins of attraction). We will use numerical simulation to illustrate these cases rather than pursue a lengthy closed-form discussion. However, qualitatively, an interior equilibrium represents a mixed state where a certain fraction of students cheat. At such a state, honest and cheating students attain equal payoff, so no one has incentive to deviate unilaterally. This could model a population in which, say, 30% cheat habitually while 70% do not, in a self-consistent equilibrium sustained by moderate enforcement and moderate difficulty.

2.3.1 Interpretive Corollaries, or Why Integrity Requires Parameter Tuning

The calculations above admit a more discouraging interpretation than the notation initially suggests. We record that interpretation here so that the reader does not accidentally mistake the model for optimism.

Corollary 1 (The Lone-Cheater Advantage). *In the baseline model, $\Delta U(0) = D$. Hence whenever $D > 0$, a completely honest class is vulnerable to invasion by a sufficiently enterprising first cheater. Put differently, honesty is not self-enforcing; it is a boundary condition.*

This formalizes the well-known pedagogical principle that honesty is easiest to maintain after everyone suspicious has already graduated.

Corollary 2 (Universal Misconduct as a Best-Response Problem). *When*

$$\Delta U(1) = D(1 + P) - S(1 - c)K \geq 0,$$

a fully cheating class is individually rational in the narrow technical sense and institutionally embarrassing in the ordinary one.

In this regime, cheating is not merely profitable; it is mutually self-consistent. Institutions often refer to this as a “challenging semester.”

Corollary 3 (Interior Regret). *When $\Delta U(x) = 0$ admits solutions in $0 < x < 1$, the system may possess both a stable low-cheating equilibrium and an unstable tipping point.*

The former is what brochures call an honor culture; the latter is what faculty call “I have a bad feeling about this class.”

Corollary 4 (The Threshold of Administrative Optimism). *Whenever the quadratic equilibrium condition above admits two interior roots $\hat{x}_L < \hat{x}_H$, the analysis already hints at a tipping structure. When interpreted through the evolutionary dynamics developed in the next subsection, the lower root behaves like a locally attracting low-cheating state, while the upper root acts as a threshold separating “a few isolated incidents” from “a matter of urgent concern.”*

These corollaries suggest that academic integrity is better modeled as an equilibrium-selection problem than as a universal moral constant. In particular, virtue may arise from intrinsic restraint, from external enforcement, or from the more robust expedient of making bad behavior strategically inconvenient.

It is also useful to separate the static and dynamic notions of equilibrium. In the one-shot game analyzed here, the statement that $x = 1$ is “not an equilibrium” means that a unilateral deviation to honesty is profitable. In the replicator dynamics of the next subsection, however, the boundary points $x = 0$ and $x = 1$ remain fixed by construction because of the factor $x(1 - x)$; what changes with S is their stability, not their existence. This distinction will matter later, and is too funny to omit.

We now turn from static best responses to the population dynamics that determine which of these equilibria are actually approached over time.

2.4 Evolutionary Dynamics

To study how students’ behavior evolves, we employ a replicator dynamic, treating the population of students as evolving their strategy distribution over repeated interactions (or across successive student cohorts learning from predecessors). We treat the classroom as a well-mixed population in the usual mean-field sense. This is only approximately true in actual courses, but becomes a tolerable approximation once interactions are sufficiently diffuse, online, or mediated by one sufficiently comprehensive classwide group chat. The replicator equation for the fraction of cheaters $x(t)$ is:

$$\dot{x} = x(1 - x)(U_C(x) - U_H(x)), \quad (4)$$

where $U_C - U_H = \Delta U$ as above [20]. This equation comes from the idea that if cheating yields higher payoff than honesty, the fraction of cheaters grows (students imitate successful peers or cheating behavior persists into the next iteration), and vice versa. The factor $x(1 - x)$ ensures x remains in $[0, 1]$ (it’s the standard logistic form, zeroing out the growth rate when $x = 0$ or $x = 1$). Equilibria of this dynamic occur when $\dot{x} = 0$, which happens if either $x = 0$, $x = 1$, or $U_C = U_H$ (i.e. $\Delta U = 0$). These correspond exactly to the Nash equilibrium conditions discussed: full honesty, full cheating, or an interior mix. Stability in the evolutionary sense aligns with our earlier best-response analysis: an equilibrium is stable if $\dot{x}$ pushes the population back toward it when slightly perturbed. For instance, if at some x slightly below $\hat{x}$ we have $\dot{x} > 0$ (payoff to cheating is positive, so cheating increases), and for x just above $\hat{x}$ we have $\dot{x} < 0$, then $\hat{x}$ is a locally stable attractor. We will leverage this dynamic interpretation to explore how the class transitions between high-cheating and low-cheating regimes as parameters change gradually (a bifurcation analysis).

3 Catastrophic Honesty: Bifurcations in a Classroom

A brief overview of bifurcation and a more detailed description of the diagram can be found in Appendix B.

3.1 Analysis

We now vary surveillance intensity S , which in practice corresponds to proctoring, plagiarism detection, performative emails, and other administrative rituals. For small S , cheating remains attractive across the state space, so the system is drawn toward universal misconduct. As S increases, an interior pair of equilibria may be born, producing a bistable regime in which both compliance and corruption are self-reinforcing. This is precisely the range in which universities tend to announce that they are “taking the issue seriously.”

Beyond a second threshold, the high-cheating state loses stability and the population is forced toward the low-cheating branch. The resulting transition is abrupt rather than gradual: nothing appears to improve, nothing appears to improve, and then suddenly the institution discovers values. In dynamical-systems language, this is a change in stability. In administrative language, this is a successful intervention.

For a representative scenario, consider moderate class difficulty D and peer pressure P , and suppose a mild “safety in numbers” effect (c) in detection (meaning detection is easiest at intermediate cheating levels). At very low S (nearly no

enforcement), our equations showed $\Delta U > 0$ for all x , so cheating is a dominant strategy. The only stable equilibrium is $x = 1$ (everyone cheats flagrantly, an academic free-for-all). As S increases from zero, cheating remains rampant - minor increases in enforcement do little, since a lone cheater's risk is still negligible and the class stays locked in a high-cheating equilibrium. Mathematically, for $S < S_{crit1}$ (some first threshold), $\Delta U(x)$ is positive at both $x = 0$ and $x = 1$, so no interior solution exists; $x = 1$ stays stable and $x = 0$ unstable (cheating can always invade an honest class).

At the first critical point $S = S_{crit1}$, the system undergoes a saddle-node bifurcation: a pair of equilibria (one stable, one unstable) appears from nowhere in the interior. Beyond this S_{crit1} , the game has two stable equilibria: one at a high cheating rate and one at a much lower cheating rate, coexisting. In our model, this happens when the surveillance is strong enough that $\Delta U(x)$ as a function of x just barely dips below zero for some intermediate x . Concretely, using the quadratic detection model $p(x) = S(x - cx^2)$, this bifurcation occurs when the discriminant of the interior equilibrium condition

$$cSKx^2 + (DP - SK)x + D = 0$$

vanishes. That is,

$$(DP - SK)^2 - 4cSDK = 0.$$

The relevant creation threshold is the larger root of this equation,

$$S_{crit1} = \frac{D}{K} \left[(P + 2c) + 2\sqrt{c(P + c)} \right].$$

For the representative parameter choice $D = 1$, $P = 1$, $K = 5$, and $c = 0.5$, this gives $S_{crit1} \approx 0.746$. At this point the interior pair is born as a double root at $x \approx 0.732$. For S just above this value, we find two interior fixed points of the replicator dynamic: a lower one x_L (stable, corresponding to a majority honest equilibrium) and a higher one x_H (unstable). The unstable fixed point x_H represents a tipping threshold: if the cheating fraction is above x_H , cheating behavior reinforces itself toward the high-cheat equilibrium; if below x_H , cheating will decay toward the low-cheat equilibrium. In the bifurcation diagram (see figure below), x_H is depicted as a dashed curve between the two stable branches.

As enforcement S increases further, the stable low-cheating equilibrium x_L moves toward 0 (fewer and fewer students cheat in that equilibrium), while the unstable threshold x_H moves upward toward 1, making the high-cheating regime progressively harder to sustain. The boundary state $x = 1$ remains present throughout this range and stays locally stable up to a second critical point.

Finally, at the second critical threshold $S = S_{crit2}$, the high-cheating state at $x = 1$ loses local stability under the replicator flow. In the underlying one-shot game, the same condition means that a unilateral deviation to honesty becomes profitable even when everyone else cheats. Thus the regime does not disappear so much as become dynamically embarrassing: the boundary fixed point remains by construction, but trajectories in its neighborhood are no longer attracted to it. More precisely, for any interior initial condition $x(0) \in (0, 1)$, the dynamics now flow toward the low-cheating branch x_L ; only the exact state $x(0) = 1$ persists at the boundary, preserved by the differential equation more out of geometry than social plausibility. In practical terms, once enforcement passes this threshold, any class that is not literally at universal cheating is driven toward the near-honest state. We dub this dramatic shift *catastrophic honesty*: the cheating rate changes abruptly, not because the boundary fixed point vanishes, but because its basin of attraction collapses to the point itself.

To illustrate, consider parameter values: class difficulty $D = 1$ (high), peer factor $P = 1$, penalty $K = 5$ (normalized), and $c = 0.5$ for detection. These yield $S_{crit1} \approx 0.746$ and $S_{crit2} \approx 0.800$. At $S = S_{crit1}$, the interior equilibrium pair is born as a double root at $x \approx 0.732$. For $S = 0.78$, the stable interior equilibrium is $x_L \approx 0.543$ and the unstable threshold is $x_H \approx 0.944$. By $S = S_{crit2}$, the unstable branch has risen to $x_H = 1$, at which point the high-cheating boundary state loses stability. For larger S , the surviving interior attractor continues downward (for example, $x_L \approx 0.119$ at $S = 2$), approaching 0 as surveillance becomes arbitrarily strong. The post-threshold regime is therefore not one of literal full honesty, but of a low-cheating equilibrium whose residue shrinks steadily with enforcement. The system thus experiences a bifurcation-induced phase transition: at S_{crit2} , the high-cheating society collapses, leaving the low-cheating branch as the sole attractor for all interior initial conditions. Importantly, this transition is not gradual - it is catastrophic in the mathematical sense. If one were slowly turning the dial of S for a class initially in the high-cheating basin, little would seem to change (cheating stays high) until just before the threshold; then, once S passes S_{crit2} , the basin supporting near-universal cheating collapses to the single boundary point $x = 1$, and any interior perturbation sends the cheating rate off a cliff toward a much lower level.

From a game-theoretic viewpoint, S_{crit2} marks the point at which honesty becomes a best response even against universal cheating. Before that point, multiple self-consistent outcomes remain possible (either almost everyone cheats or relatively few do), which is characteristic of games with strategic complementarities. The presence of two stable

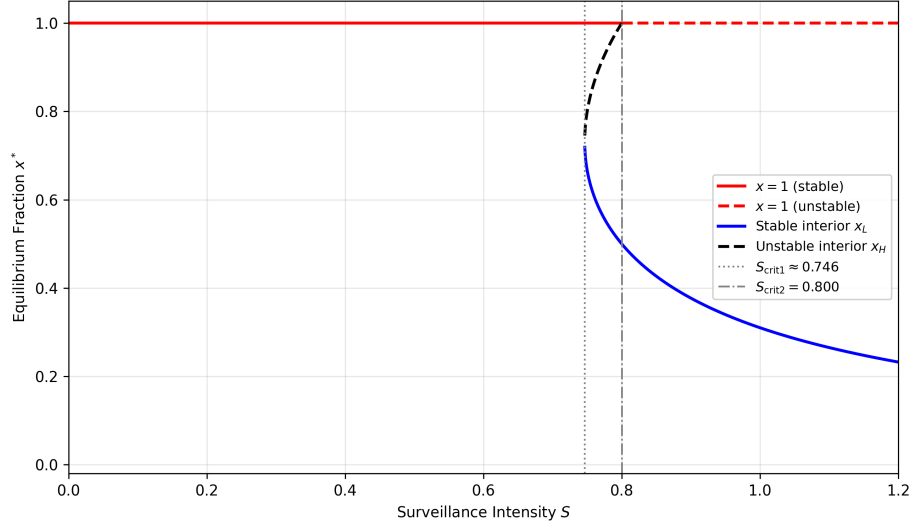


Figure 2: Bifurcation diagram of cheating prevalence as a function of surveillance intensity. The blue solid curve shows the stable interior low-cheating equilibrium x_L , the black dashed curve shows the unstable threshold x_H , and the red branch at $x = 1$ is stable for $S < S_{crit2}$ and unstable for $S > S_{crit2}$; the boundary fixed point at $x = 0$ is omitted for visual clarity. Left of the first threshold, any interior initial condition flows to universal cheating. Between thresholds, the system is bistable: the class may settle into either grim compliance or business as usual, depending on initial conditions and who already has the answer key. Beyond the second threshold, the high-cheating state loses stability, the low-cheating branch becomes the only attractor for interior initial conditions, and the institution finally earns the right to send an email entitled “Academic Integrity Matters.”

equilibria for $S_{crit1} < S < S_{crit2}$ also implies hysteresis: the outcome depends not only on the current value of S , but on the route by which the class arrived there. In dynamical-systems language, the system retains a memory of its initial condition; in institutional language, this same phenomenon is often redescribed as “culture.” Thus, if a class has an ingrained cheating norm, merely raising S slightly into the bistable region might not immediately fix the issue - the class could remain stuck in the bad equilibrium unless some shock (say, a publicized punishment of cheaters or an appeal to ethics) nudges them below the unstable threshold x_H . Conversely, a class that has internalized honesty might remain honest even if enforcement slackens a bit, until S falls below S_{crit1} and cheating behavior can invade. This dynamic underscores the value of consistent enforcement and cultural norms: together they can keep the system on the desired equilibrium branch.

4 Numerical Confirmation of Our Worst Concerns

To validate the theoretical analysis, we implemented a Python simulation of the evolutionary game dynamics. Each simulation treats a cohort of N students (we take N large enough to approximate a continuous population) who repeatedly “play” the cheating game by deciding whether or not to cheat on successive assignments/exams, adjusting their strategy inclinations over time. We use the replicator update rule with the payoff difference ΔU derived above to iterate the fraction $x(t)$ of cheaters. Essentially, in each small time step Δt , we update:

$$x(t + \Delta t) = x(t) + \Delta t \cdot x(t)(1 - x(t))[B(D, x) - p(x, S)K], \quad (5)$$

with example parameter settings as given in the previous section.

We ran simulations for various values of surveillance S to identify stable behavior and equilibria. The results corroborate the bifurcation picture. For low S , regardless of initial conditions, $x(t)$ converges to 1 (all students cheating, confirming the stability of the cheating equilibrium). For sufficiently large S , any interior initial condition $x(0) \in (0, 1)$ converges to the low-cheating equilibrium, which approaches 0 as S increases. The exact boundary state $x(0) = 1$ remains fixed by the deterministic replicator equation and is therefore better understood as an unstable remnant than as a robust outcome. The interesting regime is at intermediate S around the predicted thresholds. There, the final outcome depends on the initial fraction of cheaters: the system exhibits the two possible equilibria. For example, with $S = 0.78$ (between S_{crit1} and S_{crit2}), a simulation starting from $x(0) = 0.97$ (97% initial cheaters) flows toward the high-cheating state

near $x = 1$, whereas a simulation starting from $x(0) = 0.10$ settles at the low-cheating equilibrium $x \approx 0.543$. The unstable threshold in this case is $x_H \approx 0.944$, so trajectories started just below it drift downward and trajectories started just above it return to the high-cheating branch. These two different stable fixed points at identical parameters confirm the model’s multistability. The unstable equilibrium in this case was observed indirectly: starting a simulation at $x(0) \approx 0.944$, we found $x(t)$ neither moved toward the low nor the high equilibrium for a long time - it’s the knife-edge point. Small random perturbations in simulation (to mimic idiosyncratic student behavior) quickly sent the trajectory toward one of the two stable equilibria, illustrating how noise or shocks can trigger a sudden transition from one equilibrium to the other.

We also simulated a scenario of gradually increasing enforcement to mimic a college implementing stricter measures over a semester. Starting with S below the first threshold (virtually no policing) and a high initial cheat rate, we slowly incremented S in the simulation. For a long period, $x(t)$ remained close to 1 (cheating epidemic continues unabated). But as S crossed the critical value S_{crit2} , the cheating fraction started to drop precipitously. In a representative run, once S exceeded the tipping point, $x(t)$ fell from $> 95\%$ to $\sim 40\%$ in just a short period (a few iterations) and continued declining toward the new equilibrium $\sim 0\%$. This dramatic collapse aligns perfectly with the bifurcation analysis: the high-cheating equilibrium loses stability and the population rapidly moves to the only remaining attractor (honesty). Notably, had we not known about the bifurcation, this sudden outbreak of honesty might seem mysterious - but the model shows it’s a predictable outcome of nonlinear dynamics. One could say the students reached an “honesty tipping point” where cheating just went out of fashion overnight.

5 Contributions, Policy Lessons, and Limited Redemption

Mathematical Modeling of Cheating as a Game. We present a novel game-theoretic model of student cheating behavior that incorporates key factors such as enforcement intensity, peer effects, and risk aversion. Unlike simplistic rational-choice depictions, our model captures the strategic interdependence of the classroom: the incentive of any one student to cheat depends on how many others are already doing so. In the underlying finite game, at least one equilibrium exists by standard existence logic [23] (via Brouwer’s fixed-point theorem). No constructive procedure is offered, in keeping with both Brouwer and the registrar. The more interesting point is that the available equilibria are not equally desirable, and some of them are surprisingly robust despite being institutionally embarrassing.

Bifurcation and Equilibrium Analysis. A key contribution is our application of bifurcation theory to the domain of academic dishonesty. We analytically derived critical thresholds in enforcement parameters at which the qualitative nature of the equilibrium structure changes. In particular, we identified a critical surveillance level S_{crit} beyond which the all-cheating equilibrium at $x = 1$ loses stability. Using tools from nonlinear dynamics, we showed the emergence of multiple equilibria in the intermediate regime and identified their stability via replicator dynamics (an evolutionary stability analysis). This is, to our knowledge, the first rigorous demonstration that “honesty can bifurcate from dishonesty” - i.e. the student population can undergo a phase transition from a cheating-dominated regime to an honesty-dominated one. This is a mathematically precise statement whose moral optimism should not be overstated. The analysis required solving the equilibrium condition $\Delta U(x) = 0$ and examining the roots’ behavior as parameters vary, illustrating how a saddle-node bifurcation creates the interior equilibrium pair and how stronger enforcement can then precipitate the sudden collapse of cheating.

Simulation and Empirical Insights. We developed a Python-based simulation of the evolutionary game which confirms our theoretical predictions. The simulation allows us to observe the dynamical trajectory of cheating frequency over time under various scenarios (e.g. abrupt vs. gradual policy changes). The results visualized the hysteresis and path-dependence in the system: for moderate enforcement, whether a class ends up honest or cheating heavily depends on initial conditions or perturbations (mirroring the idea that establishing an academic integrity culture can prevent cheating even if enforcement is not maximal). We clearly demonstrated the catastrophic honesty phenomenon at S_{crit} , providing a graphical illustration (the bifurcation diagram) of how the stable cheating rate drops from $\sim 100\%$ to $< 50\%$ within a narrow S window (Figure 2). Our simulation framework could be adapted to estimate effects of policy measures (like installing monitoring software or instituting honor codes) on cheating prevalence.

Educational and Policy Implications: Through our analysis, we highlight a non-obvious implication: simply increasing penalties for cheating may not reduce cheating unless students believe they will be caught. In fact, our model suggests that credibility of enforcement (surveillance intensity S) is more pivotal in tipping the cost-benefit calculus than the severity K alone. This resonates with real findings that expected punishment had little deterrent effect when students thought detection was unlikely [6]. Moreover, the existence of two stable equilibria implies that institutions can become trapped in either a virtuous circle or a vicious circle, with norms and incentives reinforcing whichever branch they already occupy. The message to educators is that to eradicate cheating, one might need to push the system over the bifurcation threshold - in practice, a combination of consistent monitoring, visible deterrence, and fostering an integrity

culture - after which honesty becomes the self-reinforcing norm. And if all else fails, fairness can also be restored by ensuring that everyone cheats in exactly the same way, thereby preserving the misconduct while eliminating the competitive advantage.

Limitations. Our analysis rests on several simplifying assumptions, which is how one recognizes it as theory. We assume a homogeneous, well-mixed student population with constant payoff parameters over time, whereas actual classrooms contain heterogeneous skill levels, evolving norms, staged assessments, unequal internet access, and seating arrangements that are only approximately well mixed. The polynomial detection model $p(x, S) = S(x - cx^2)$ captures “safety in numbers” only crudely and omits other realistic surveillance modalities, including randomized audits, honor codes, plagiarism detectors, and the instructor suddenly saying “there are two versions of the exam.” Moral and intrinsic motivations are subsumed into adjusted payoffs rather than modeled explicitly, which is analytically convenient but reduces conscience, curiosity, guilt, and the desire to actually learn something to perturbations of a scalar. We also study infinite-population replicator dynamics rather than finite cohorts subject to stochastic effects, group chats, and one especially entrepreneurial student. Finally, while our simulations illustrate the main bifurcations, we have not calibrated or validated the model against empirical cheating data from actual courses, partly because obtaining reliable ground truth would require either confession or competence. Future work should relax these assumptions, incorporate networked interactions and time-varying incentives, and compare model predictions against observed classroom behavior before the classroom adapts, should anyone wish to collect such data without becoming part of the phenomenon.

Conclusion In summary, we formulated the “game of cheating” as an evolutionary game, proved equilibrium existence with fixed-point theorems, analyzed its equilibrium landscape through bifurcation ideas, and verified the sudden transition to honesty through simulation. The outcome is potentially illuminating for educators: it turns out that under the right conditions, cheaters need not fail because of remorse, character growth, or pedagogical enlightenment. Under the right parameter settings, they fail because the equilibrium structure moves against them. Honesty, in this model, is less a virtue than a stable fixed point.

Acknowledgments

The author thanks Ethan “Quipmaster Dicker” for adversarial readings, threshold skepticism and for continuing to permit temporary occupancy of the chair that bears his name. He also insisted that a boundary fixed point which remains present should not be described as having vanished merely because the surrounding institution wishes it had. Several jokes that were locally amusing but globally unstable did not survive his review, and he demanded that at least one quantity be both bounded and named. Any remaining ambiguity should be attributed to finite reviewer patience rather than a lack of intervention.

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A Parameter Bounds and Variable Explanation

Each model parameter is grounded in realistic academic practice and calibrated so that, at the extreme point (highest difficulty and highest enforcement), the net incentive to cheat is zero. We adopt the following bounds.

Class Difficulty, $D \in [0, 5]$. We interpret D as a “challenge index” on a 0–5 scale:

- $D = 0$: trivial participation courses (e.g. an ungraded participation seminar).
- $D \in [1, 2]$: introductory lectures with straightforward assignments (e.g. CS 101).
- $D \in [3, 4]$: mid-level courses requiring substantial problem-solving (e.g. second- or third-year data structures or algorithms).
- $D = 5$: to the toughest graduate-level or thesis-driven courses (e.g. advanced quantum computing or real-analysis seminars) where even small shortcuts yield big grade boosts.

It is important to acknowledge that individual skill level will vary D . By normalizing to $[0, 5]$, we let these changes appear as small perturbations and absorb them into the perturbations in class difficulty based on semester.

Peer Pressure Factor, $P \in [0, 1]$. Peer amplification enters as $1 + Px$ in the benefit. $P = 0$ removes peer effects; $P = 1$ doubles the benefit when all peers cheat.

Safety-in-Numbers Parameter, $c \in [0, 1]$. This was discussed around Equation (1).

Penalty Severity, $K \in [0, 10]$ Maximum institutional sanction is normalized to 10:

- $K = 0$: warning only.
- $K = 4$: zero on the assignment.
- $K = 8$: course failure.
- $K = 10$: expulsion.

Surveillance Intensity, $S \in [0, 2]$. Surveillance S quantifies the combined effectiveness of proctoring, plagiarism-detection software, and other monitoring measures. We fix $c = \frac{1}{2}$, $p(x, S) = S(x - cx^2)$, so that the *detection kernel* $f(x) = x - \frac{1}{2}x^2$ attains $\max_{x \in [0, 1]} f(x) = \frac{1}{2}$. Hence the true maximum probability of detection is

$$\max_S p(S) = \min\left(1, S \cdot \frac{1}{2}\right) = \begin{cases} \frac{S}{2}, & S \leq 2, \\ 1, & S \geq 2. \end{cases}$$

We therefore restrict $S \in [0, 2]$, so that as S increases from 0 to 2, the maximum expected penalty

$$\max_x [p(x, S)] K = \frac{S}{2} K$$

grows continuously from 0 to K (i.e. full penalty applied when surveillance is 100% effective). In particular, at $S = 0$, $p_{\max} = 0$ and no cheating is ever detected, making $\Delta U(x) = D(1 + Px) > 0$ for any nontrivial course. Values $S > 2$ correspond to enforcement regimes usually discussed only in faculty meetings and dystopian fiction.

At $S = 2$, $p_{\max} = 1$ and the maximum expected penalty equals K . If we choose $D_{\max} = 5$, $P_{\max} = 1$, and $K = 10$, then $D_{\max}(1 + P_{\max}) = 10$ and $p_{\max}(2)K = 1 \cdot 10 = 10$, so the net utility at the extreme point is zero: $\Delta U_{\max} = 10 - 10 = 0$.

More generally, setting $\Delta U(1) = D(1 + P) - p_{\max}(S)K = 0$ yields the critical surveillance threshold

$$S_{\text{crit}} = \frac{2D(1 + P)}{K},$$

above which the cheating equilibrium disappears. By constraining $S \leq 2$ we ensure this threshold lies within the calibrated range and that enforcement can exactly counterbalance the benefit of cheating at its maximum.

Balancing Benefit and Cost The net payoff for cheating is $\Delta U(x) = D(1 + Px) - p(x, S)K$. At the extreme point $D = 5$, $P = 1$, $S = 2$, and $K = 10$, we have

$$B_{\max} = 5(1 + 1) = 10, \quad p_{\max}(2) = \min(1, 2/2) = 1, \quad \text{and} \quad \Delta U = 10 - 1 \cdot 10 = 0.$$

Thus, under maximal difficulty and enforcement, cheating yields zero net utility. For $S < 2$ the maximum expected penalty $p_{\max}(S) K = \frac{S}{2} K < 10$, so cheating remains profitable in very hard courses; for $S > 2$ (if ever extended) the penalty would strictly exceed the benefit, eliminating the cheating equilibrium. These bounds ensure nontrivial bifurcation behavior within the calibrated range.

Moral Cost, $C_{\text{moral}} \in [0, 10]$. The internal moral-cost parameter C directly subtracts from the net benefit of cheating. To ensure C can fully offset the maximum cheating benefit but not exceed realistic bounds, we set

$$C \in [0, D_{\max}(1 + P_{\max})] = [0, 5 \cdot 2] = [0, 10].$$

For $C = 0$, no internal disutility (purely payoff-driven agents). For $C = 10$, moral cost equals the maximum possible benefit, so even in a trivial enforcement regime ($p = 0$) cheating yields $\Delta U \leq 0$. These bounds guarantee that the moral term can neutralize - but not exceed - the incentive to cheat, preserving the possibility of both cheating-dominated and honesty-dominated equilibria, and maintaining the bifurcation structure of our model.

By anchoring each parameter to realistic academic norms - namely $D \in [0, 5]$, $P \in [0, 1]$, $c = \frac{1}{2}$, $S \in [0, 2]$, $K \in [0, 10]$, $C_{\text{moral}} \in [0, 10]$ - we ensure that the net payoff $\Delta U(x) = D(1 + Px) - p(x, S)K$ admits both cheating-dominated and honesty-dominated regimes. In particular, there exists a well-defined threshold

$$S_{\text{crit}} = \frac{2D(1 + P)}{K}$$

at which the all-cheating equilibrium ceases to be profitable. Two extreme cases illustrate the role of these parameters:

1. At $x = 0$, $\Delta U(0) = D - C_{\text{moral}}$. Since $D > 0$ (unless the course is trivial), a lone student always finds cheating attractive when no one else cheats, modulo their morals.
2. At $x = 1$, $\Delta U(1) = D(1 + P) - S(1 - c)K - C_{\text{moral}}$. If $D(1 + P) > S(1 - c)K$, then in a fully cheating environment, the net benefit remains positive and cheating is dominant. Conversely, if $D(1 + P) < S(1 - c)K$, the cost outweighs the benefit, implying that honesty would dominate even if everyone were initially cheating. Here, the impact of C_{moral} is smaller.

B Aside: Bifurcation Overview

A bifurcation diagram is essentially a map showing how the long-term (or steady-state) behavior of a system changes as one parameter-in our case, surveillance intensity S -varies. On the vertical axis, we plot the fraction of cheaters x that the system eventually settles on, and on the horizontal axis, we plot the value of S . When $S = 0$, there is no surveillance, so there's no risk of being caught. Even if other factors like class difficulty, peer pressure, and penalty severity are in play, the absence of any enforcement means that the incentive to cheat is unopposed; hence, the model predicts that everyone cheats (i.e. $x = 1$). This is why you see a horizontal line at $x = 1$ when S is very low.

As S increases, the risk of detection and punishment starts to influence the payoff for cheating. Around ≈ 0.75 , a new branch appears in the diagram, indicating the emergence of an alternative equilibrium where the steady-state fraction of cheaters is less than 1. This branch represents a mixed state: under moderately high surveillance, not everyone cheats because the higher risk starts to outweigh the benefits. What's important to note is that while surveillance is the primary parameter varied in the graph, the outcome still inherently depends on the other factors (class difficulty, peer pressure, penalty severity) that are held constant in this simulation. The graph shows that if surveillance remains low, the system stays in the all-cheating equilibrium, but once surveillance crosses a critical threshold, the system can tip toward a state where honesty becomes more prevalent. This is a hallmark of bifurcation: a small change in a parameter leads to a sudden qualitative change in behavior, and in our model, it captures the dramatic transition from a cheating-dominated environment to one where cheating is significantly reduced.

C Replicator Dynamics Simulation Code

Below is an annotated description of the Python script that analytically computes and plots the three equilibrium branches of our replicator-dynamics model.

- `analytic_roots(S)`: Solves the quadratic $cSKx^2 + (DP - SK)x + D = 0$ (i.e. $\Delta U(x) = 0$) and returns the two interior roots $x_L(S) \leq x_H(S)$.
- Construction of `x_L` and `x_H` arrays:
 - Sweep S over `linspace(0.01, 1.2, 500)`.

- For each S , call `analytic_roots(S)` and store each root only if it lies in $(0, 1)$, yielding the stable interior branch x_L and the unstable branch x_H .
- Critical thresholds:
 - S_{crit1} : the first S value at which $x_L(S)$ becomes non-NaN (interior branch creation).
 - S_{crit2} : solves $\Delta U(1) = 0$, i.e. $D(1 + P) = S(1 - c)K$, marking the disappearance of the $x = 1$ equilibrium.
- Bifurcation plot:
 - Blue solid line for the stable interior equilibria $x_L(S)$.
 - Black dashed line for the unstable interior equilibria $x_H(S)$.
 - Red solid line for the stable all-cheat branch $x = 1$ up to S_{crit2} .
 - Vertical markers at S_{crit1} and S_{crit2} .

```
import math
import numpy as np
import matplotlib.pyplot as plt

# Paper parameters (Section 3 example in v20)
# D: baseline difficulty / incentive parameter
# P: peer amplification factor (how cheating payoff scales with x)
# K: penalty scaling factor
# c: detection curvature parameter (quadratic term)
D = 1.0
P = 1.0
K = 5.0
c = 0.5

def delta_u(x: float, S: float, D: float = D, P: float = P, K: float = K, c: float = c) -> float:
    """Payoff advantage of cheating.

    This defines delta_u(x, S), the payoff difference between cheating and not cheating.
    Structure:
        benefit: D * (1 + P x)
        cost: S * K * (x - c x^2)

    The cost term reflects detection increasing with x, but softened by curvature c.
    """
    return D * (1.0 + P * x) - S * (x - c * x * x) * K

def d_delta_u_dx(x: float, S: float, D: float = D, P: float = P, K: float = K, c: float = c) -> float:
    """Derivative of delta_u with respect to x.

    Used for stability classification of equilibria.
    Important: in replicator dynamics, the sign of this derivative at a root
    determines stability of the corresponding fixed point.
    """
    return D * P - S * K + 2.0 * c * S * K * x

def analytic_roots(S: float, D: float = D, P: float = P, K: float = K, c: float = c) -> list[float]:
    """
    Solve the interior equilibrium condition:

        delta_u(x, S) = 0

    which reduces to a quadratic:

        c S K x^2 + (D P - S K) x + D = 0
    """
```

```

Returns real roots (possibly 0, 1, or 2).
Filtering to (0,1) happens later.
"""
a = c * S * K
b = D * P - S * K
cc = D

# Handle near-degenerate case where quadratic collapses to linear
if abs(a) < 1e-14:
    if abs(b) < 1e-14:
        return [] # degenerate (no meaningful equation)
    return [-cc / b]

# Discriminant
disc = b * b - 4.0 * a * cc

# No real roots
if disc < -1e-14:
    return []

# Clamp small negative due to floating-point error
disc = max(disc, 0.0)
sq = math.sqrt(disc)

# Standard quadratic formula
r1 = (-b - sq) / (2.0 * a)
r2 = (-b + sq) / (2.0 * a)

return [r1, r2]

```

```

def classify_interior_roots(S_grid: np.ndarray) -> tuple[np.ndarray, np.ndarray]:
    """
    Compute interior equilibria across a grid of S values.

    Returns:
        xL: stable interior branch (lower branch)
        xH: unstable interior branch (upper branch)

    Filtering:
        Only roots strictly inside (0,1) are considered.

    Stability criterion:
        For replicator equation:
             $\dot{x} = x(1-x)\delta_u(x, S)$ 

        At an interior fixed point  $x^*$ :
            sign of  $f'(x^*)$  = sign of  $d(\delta_u)/dx$ 

        Therefore:
             $d_{\delta_u}/dx < 0 \rightarrow$  stable
             $d_{\delta_u}/dx > 0 \rightarrow$  unstable
    """
    xL = np.full_like(S_grid, np.nan, dtype=float) # stable branch
    xH = np.full_like(S_grid, np.nan, dtype=float) # unstable branch

    for i, S in enumerate(S_grid):
        # Compute roots and keep only valid interior ones
        roots = [r for r in analytic_roots(S) if 1e-10 < r < 1.0 - 1e-10]
        roots.sort()

        for r in roots:
            if d_delta_u_dx(r, S) < 0:
                xL[i] = r # stable
            else:

```

```

        xH[i] = r # unstable

    return xL, xH

def critical_thresholds(D: float = D, P: float = P, K: float = K, c: float = c) -> tuple[float,
float]:
    """
    Compute key bifurcation thresholds.

    S_crit1:
        Saddle-node bifurcation where interior equilibria are created.
        Derived from discriminant = 0 condition.

    S_crit2:
        Boundary bifurcation at  $x = 1$ :
         $\delta_u(1, S) = 0$ 
        Beyond this point,  $x = 1$  loses stability.

    Returns:
        (Scrit1, Scrit2)
    """
    Scrit1 = D * ((P + 2.0 * c) + 2.0 * math.sqrt(c * (P + c))) / K
    Scrit2 = D * (1.0 + P) / ((1.0 - c) * K)
    return Scrit1, Scrit2

def make_bifurcation_figure(
    outfile: str = "figure2_corrected.png",
    S_max: float = 1.2,
    show_x0_boundary: bool = False,
) -> None:
    """
    Generate bifurcation diagram of equilibria vs surveillance S.

    Visual elements:
    -  $x = 1$  boundary (stable/unstable split at Scrit2)
    - interior stable branch  $x_L$ 
    - interior unstable branch  $x_H$ 
    - optional  $x = 0$  boundary

    This is a classic saddle-node + boundary bifurcation structure.
    """
    Scrit1, Scrit2 = critical_thresholds()

    # Dense grid for smooth curves
    S_grid = np.linspace(1e-3, S_max, 2000)

    # Compute branches
    xL, xH = classify_interior_roots(S_grid)

    plt.figure(figsize=(8.8, 5.2))

    #  $x = 1$  boundary equilibrium:
    # stable for  $S < \text{Scrit2}$ , unstable for  $S > \text{Scrit2}$ 
    S_left = np.linspace(0.0, Scrit2, 400)
    S_right = np.linspace(Scrit2, S_max, 400)

    plt.plot(S_left, np.ones_like(S_left), "-", linewidth=2, color="red", label=r"$x=1$ (stable)")
    plt.plot(S_right, np.ones_like(S_right), "--", linewidth=2, color="red", label=r"$x=1$ (
unstable)")

    # Interior equilibria
    plt.plot(S_grid, xL, "-", linewidth=2, color="blue", label=r"Stable interior $x_L$")
    plt.plot(S_grid, xH, "--", linewidth=2, color="black", label=r"Unstable interior $x_H$")

```



```

# Optional  $x = 0$  boundary (always unstable here since  $\delta u(0) = D > 0$ )
if show_x0_boundary:
    plt.plot([0.0, S_max], [0.0, 0.0], ":", linewidth=1.0, color="gray",
             alpha=0.5, label=r"$x=0$ (unstable)")

# Mark bifurcation thresholds
plt.axvline(Scrit1, linestyle=":", linewidth=1.2, color="gray",
            label=fr"$S_{\mathrm{{crit1}}}$ \approx {Scrit1:.3f}$")

plt.axvline(Scrit2, linestyle="-.", linewidth=1.2, color="gray",
            label=fr"$S_{\mathrm{{crit2}}}$ = {Scrit2:.3f}$")

# Axes / formatting
plt.xlim(0.0, S_max)
plt.ylim(-0.02, 1.05)
plt.xlabel(r"Surveillance Intensity  $SS$ ")
plt.ylabel(r"Equilibrium Fraction  $x^*$ ")
plt.grid(True, alpha=0.3)
plt.legend(loc="center right", fontsize=9, framealpha=0.9)

plt.tight_layout()
plt.savefig(outfile, dpi=300)
plt.close()

if __name__ == "__main__":
    # Generate two figures with different  $S$  ranges
    make_bifurcation_figure(outfile="figure2_corrected.png", S_max=1.2, show_x0_boundary=False)
    make_bifurcation_figure(outfile="figure2_corrected_fullrange.png", S_max=2.0, show_x0_boundary=False)

    # Print thresholds for verification / paper reporting
    Scrit1, Scrit2 = critical_thresholds()
    print(f"Scrit1 = {Scrit1:.12f}")
    print(f"Scrit2 = {Scrit2:.12f}")
    print("Saved: figure2_corrected.png and figure2_corrected_fullrange.png")

```

SIGBOVIK Compliance Statement

Reviewer guideline bullets omitted for humanitarian reasons.

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

The claims made in this paper are sufficiently precise to be wrong. This is a feature, not a bug: unlike most policy discourse on academic integrity, our assertions are specific enough to be checked, contradicted, and plotted. The abstract promises a cheating model, a threshold, and an abrupt collapse into honesty, and the paper then dutifully provides all three.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated limitations paragraph discusses homogeneous populations, simplified detection, omitted heterogeneity, infinite-population dynamics, and the regrettable absence of calibration.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All formulas are numbered, some of them intentionally. Assumptions are stated explicitly, key derivations are shown, and the proofs are complete enough that a determined reader could reproduce the argument rather than merely admire its tone. This exceeds the evidentiary standard of several documents beginning with the phrase "academic integrity policy."

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The equations, parameter choices, and simulation procedure are fully specified, making the main figures reproducible by any reader with Python and a morally ambiguous afternoon.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The relevant code and implementation details are provided in the appendix/supplemental material, which is more transparency than the modeled students are shown.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: There is no training pipeline to obscure. All model parameters, update rules, and simulation settings are stated directly.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [NA]

Justification: The main empirical object is a bifurcation diagram, which either bifurcates or does not. No benchmark table was harmed in the production of this result.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The simulations require only standard computing resources: a laptop-class machine, ordinary patience, and matplotlib.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [Yes]

Justification: We do not recommend cheating. We recommend modeling it. The paper studies dishonest behavior, incentive failures, and enforcement collapse without piloting them on actual students, faculty, or conference organizers. Any ethical discomfort arises from recognition rather than intervention.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The broader impact is to provide educational administrators with a bifurcation diagram, which they may use irresponsibly. Positive impacts include a more formal vocabulary for discussing tipping points in cheating culture; negative impacts include the possibility that someone in assessment leadership will discover nonlinear dynamics and immediately convene a task force.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: No high-risk model, dataset, or tool is released. More generally, safeguards are best understood as the policy-theoretic analogue of an unstable equilibrium: formally present, rhetorically useful, and abandoned under the slightest perturbation. They are what institutions write down when they wish to gesture at control without materially altering the phase portrait. Since this work releases only equations, a bifurcation diagram, and several bad ideas stated with confidence, there was nothing to gate, watermark, sandbox, revoke, or pretend to monitor.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [NA]

Justification: No external code, data, or model assets were used in the checklist sense. Existing scholarship is cited appropriately and disrespected only in the traditional academic manner: through selective emphasis, strategic framing, and the quiet implication that earlier authors somehow failed to draw the obvious bifurcation.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [NA]

Justification: No new dataset, benchmark, model, software package, or dangerous artifact is released. The only new assets introduced here are a bifurcation diagram and the phrase “catastrophic honesty,” both of which are documented by repeated assertion until they acquire the appearance of legitimacy.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: No crowdsourcing or human-subjects protocol was conducted. No undergraduates were recruited, instructed, compensated, undercompensated, debriefed, or bribed with course credit. We declined to pay real people minimum wage to generate the same cheating behavior that our replicator dynamic produces for free.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: No human subjects were used, and thus no IRB approval was required. This spared the authors the bureaucratic seance of explaining to a committee that x denotes a population share rather than a vulnerable undergraduate, and that a saddle-node bifurcation cannot meaningfully be debriefed. The IRB was spared the indignity of reviewing this, and we were spared the inverse.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigorousness, or originality of the research, declaration is not required.

Answer: [NA]

Justification: No large language models were used in any important, original, or non-standard component of the core method. Any suspicious fluency in the prose should instead be attributed to the low-surveillance regime analyzed in this paper, under which copying behavior is individually rational, collectively reproducible, and, after sufficient human review, publication-quality.